V-19—Measurement Canada policy on use of automatic temperature compensation

Date: 2005-07-01

Bulletin: V-19 (rev.1)

Category: Volume

Documents: Policy on Use of Automatic Temperature Compensation

Supercedes: V-19

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1.0 Purpose

The purpose of this bulletin is to describe the Measurement Canada policy for the use of automatic temperature compensation (ATC) for the measurement of liquid petroleum products.

2.0 Scope

This policy is applicable to all liquid meter applications where the automatic temperature compensation feature is used for trade purposes. Meter applications include fuel dispensers, high speed refuellers, cabinet and stationary aviation meters, vehicle mounted meter systems, and loading rack meters. This policy does not apply to the metering of liquids, such as propane, that have a vapour pressure of greater than 101.325 kPa at metering conditions.

3.0 Background

In the past few years, the petroleum industry has been converting their measuring equipment to account for the temperature of the product as it is being metered. This measurement practice is commonly referred to as automatic temperature compensation (ATC) and uses the internationally accepted reference temperature of 15 °C for volume correction. Many traders have chosen to convert all meters at their establishment to ATC at the same time, while some have chosen to convert only selected meters or product lines.

The *Weights and Measures Act* or *Regulations* do not address the issue of partial conversions, only requiring that when a device is temperature corrected, that it be appropriately marked. Although most traders have converted all devices at the same time, some traders, such as those operating retail fuel stations and vehicle mounted measuring systems, are

choosing to phase-in the cost of ATC conversion over a number of years. For example, only the higher volume products such as the regular no lead meters are initially converted to ATC, leaving the mid grade and premium products to a later date. Additionally, some traders have indicated their intention to turn the ATC function off during the warmer months and on again in the colder months. These situations highlight the need for an ATC policy in the petroleum industry to ensure that the conversion to a temperature compensated measuring system will be equitable to all parties in a trade transaction.

4.0 Policy

4.1 Partial conversions of meters

4.1.1 Volume measuring devices located at a trader's petroleum establishment (the address where the device is installed) will be categorized into different trade levels, namely: fuel dispensers, high speed refuellers, barrel fill meters, vehicle mounted meters and loading-rack meters.

4.1.2 An individual blend or grade of product will be defined as: regular no lead, mid grade, premium gasoline, diesel, aviation gas, Jet A and B fuel, furnace oil, stove oil, kerosene, and dyed oils.

4.1.3 A trader may partially convert meters to incorporate ATC at their establishment, provided that all meters for the same grade/blend of fuel within the same trade level are converted, and the ATC activated at the same time.

4.1.4 The words "Volume Corrected to 15 °C" or "Volume corrigé à 15 °C" shall be marked immediately adjacent to all visual or printed registrations of net quantity.

4.1.5 Vehicles that have only one meter for measuring multiple products must have all products temperature compensated as required by section 21(d)(ii) of the *Weights and Measures Regulations*.

4.1.6 Vehicles having more than one meter may convert selected meters as long as the corresponding register and meter ticket clearly indicate that the meter is used for temperature compensation, eg. truck with two meters - oil and gas - the oil meter has an electronic register and is ATC; the gas meter has a mechanical register and is not temperature compensated. All vehicles in the fleet, including spare vehicles, must be converted at the same time in accordance with section 4.1.3 above.

4.2 Turning off the ATC

4.2.1 Notwithstanding sales under contract, such as at loading rack terminals, a measuring device equipped with an automatic temperature compensator shall be connected, operable, and in use at all times.

4.2.2 The appropriate use of a device in trade has always been the responsibility of the trader. Section 23(a) of the Act makes it an offence for a trader who uses a device in trade for any purpose or in any manner that is prohibited in the certificate of inspection issued at the time of the most recent inspection.

4.2.3 Section 5(2) of the Regulations requires the trader to comply with section 8(b) of the Act, ie., inspection before use in trade, when a measuring device is altered to operate with automatic temperature compensation. As this would be the most recent inspection made under the Act, section 23(a) of the Act will be used to prohibit the use of the device in a manner, other than in the temperature compensated mode, once it has been initially inspected in this mode.

4.2.4 Measurement Canada inspectors an authorized service providers conducting inspections will annotate the comment section of the inspection certificate with the wording below to advise the trader that the ATC function cannot be turned off without the permission of a Measurement Canada inspector.

The device(s) described above are not to be used in trade when the automatic temperature compensation feature is not in use.

4.2.5 As per section 41 of the *Weights and Measures Regulations*, the trader and/or his service agent must contact the local Measurement Canada office when switching the measuring device back to ambient measurement. If the trader wants to turn on the ATC again, appropriate fees and charges shall be applied per section 5(2) of the Regulations.

5.0 Enforcement action

5.1 Measurement Canada officials will advise traders that are found to be in non-compliance with the above policy. It may often be appropriate to meet with non-complying traders to ensure that they fully understand the problem and the different options available to them. Continued non-compliance with this policy will result in a normally progressive enforcement action up to and including prosecution action.

5.2 The above policies will ensure that the conversion to automatic temperature compensation measurement continues to proceed in an effective and efficient manner. Measurement Canada will continue to monitor the voluntary conversion to ATC measurement systems in the petroleum industry in Canada and work with the Legal Services Branch of Industry Canada to identify the need for regulatory intervention as required.

6.0 Revisions

Revision 1, May 13, 2005, was made to make this bulletin applicable to both Measurement Canada inspectors and authorized service providers (ASPs). Some non significant format changes were also made. Also in section 2, 103 was changed to 101.325.

Date modified:
2019-12-13